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Evaluating the impact of intra-plate double subduction on global-scale plate kinematics within spherical numerical models

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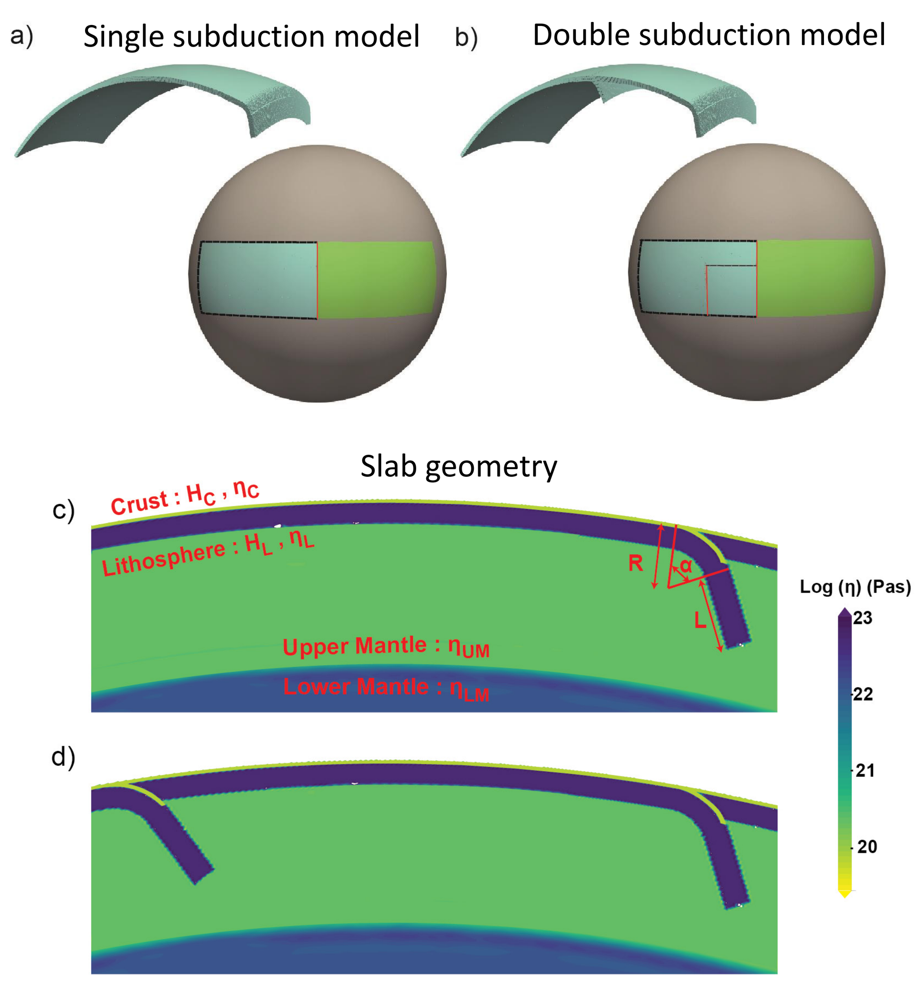
Motivation

A double subduction zone is a geodynamic setting where two subduction zones interact, affecting plate kinematics. Recent studies have focused on regional examples, like how the initiation of the Ryukyu subduction zone has altered Izu-Bonin trench motion[1,2]. However, the impact of double subduction zones on the motions of large plates, and hence global-scale plate kinematics, is less understood. It has recently been suggested that changes in Pacific Plate motion in Eocene (~50Ma)[3] and Africa Plate in the Cretaceous (108-92Ma)[4,5] are linked to the establishment of large-scale double subduction. A quantitative understanding of under which mechanical and geometrical conditions this can occur is required to evaluate the geodynamic feasibility of such suggestions.

We use 3-D numerical models to explore how the plate velocity of a large tectonic plate differs with and without the presence of a 2nd intraplate subduction zone. We map out the magnitude and direction of double-subduction induced plate velocity changes for a range of plate widths, crust and lithosphere strengths, and with/without power law flow.

Model Setup

- 3-D, spherical shell domains using ASPECT [6-9]. Geodynamic World Builder used to construct the plate and slab geometries[10]. Our models are "instantaneous" and compositional.
- Four parameters are investigated: lithosphere/mantle viscosity ratio (η_L/η_m), crust/mantle viscosity ratio (η_C/η_m), plate width (W_T) and the transition strain rate of power-law viscosity (ε_T).
- We analyze pairs of single (a,c) and double (b,d) subduction models, each with the same mechanical properties.



Single vs. Double plate velocities (reference models)

Our reference models have η_L/η_m = 200, η_C/η_m = 0.5, W_T = 30° (~3300 km), and a linear viscous rheology. Shown below are the plate velocities for both models (a,b). Also shown is the ratio between the velocity magnitudes (c) and the azimuthal difference (d) for single vs. double slab models.

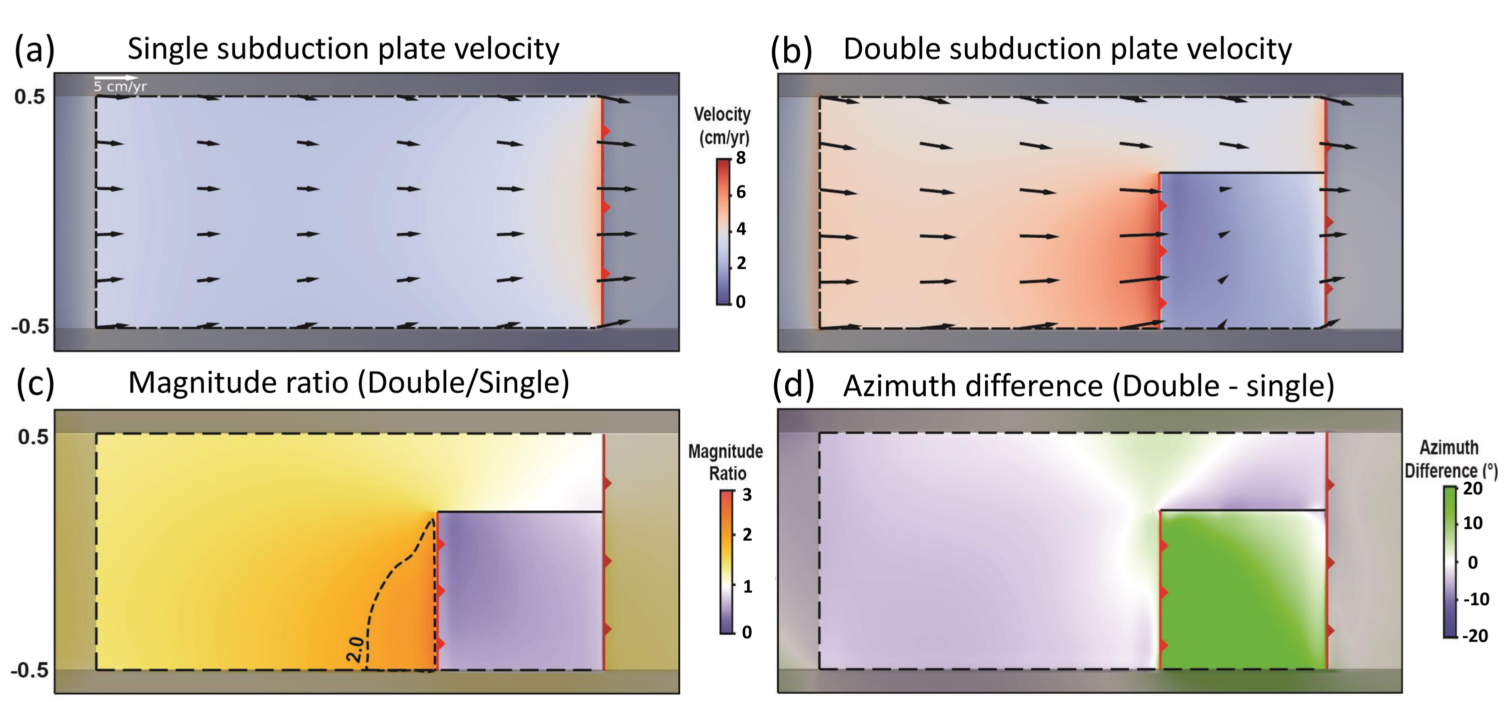
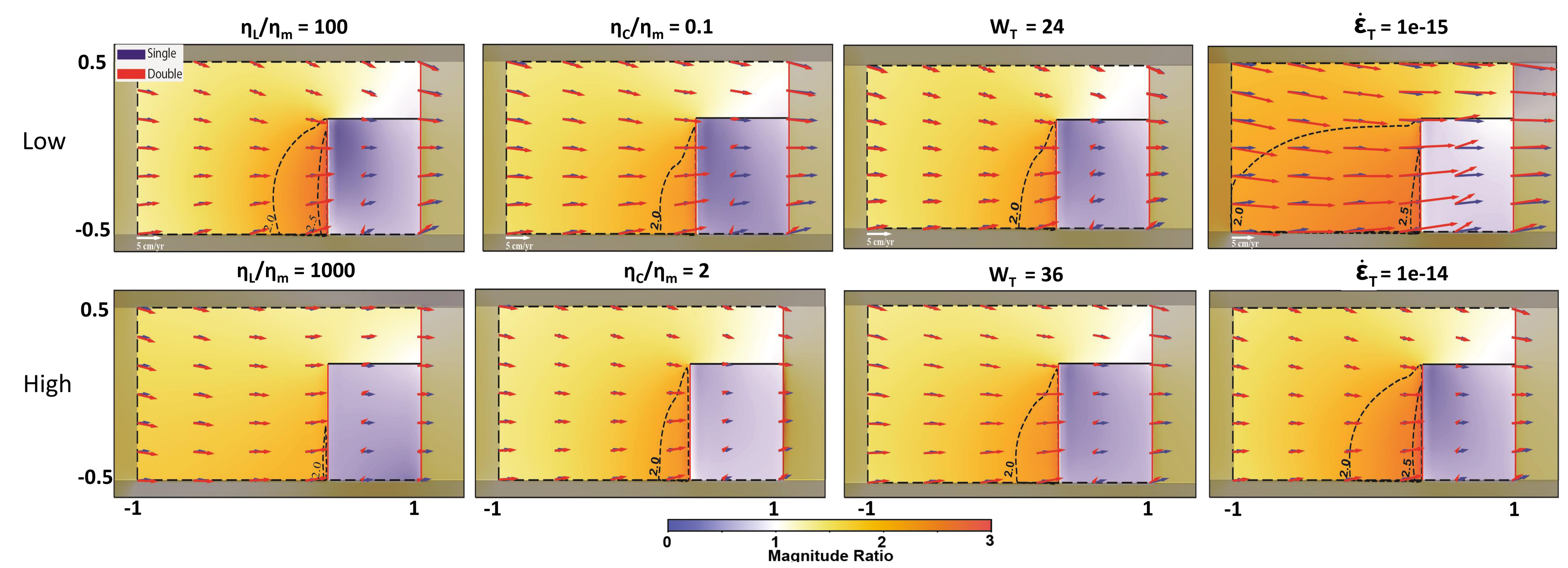
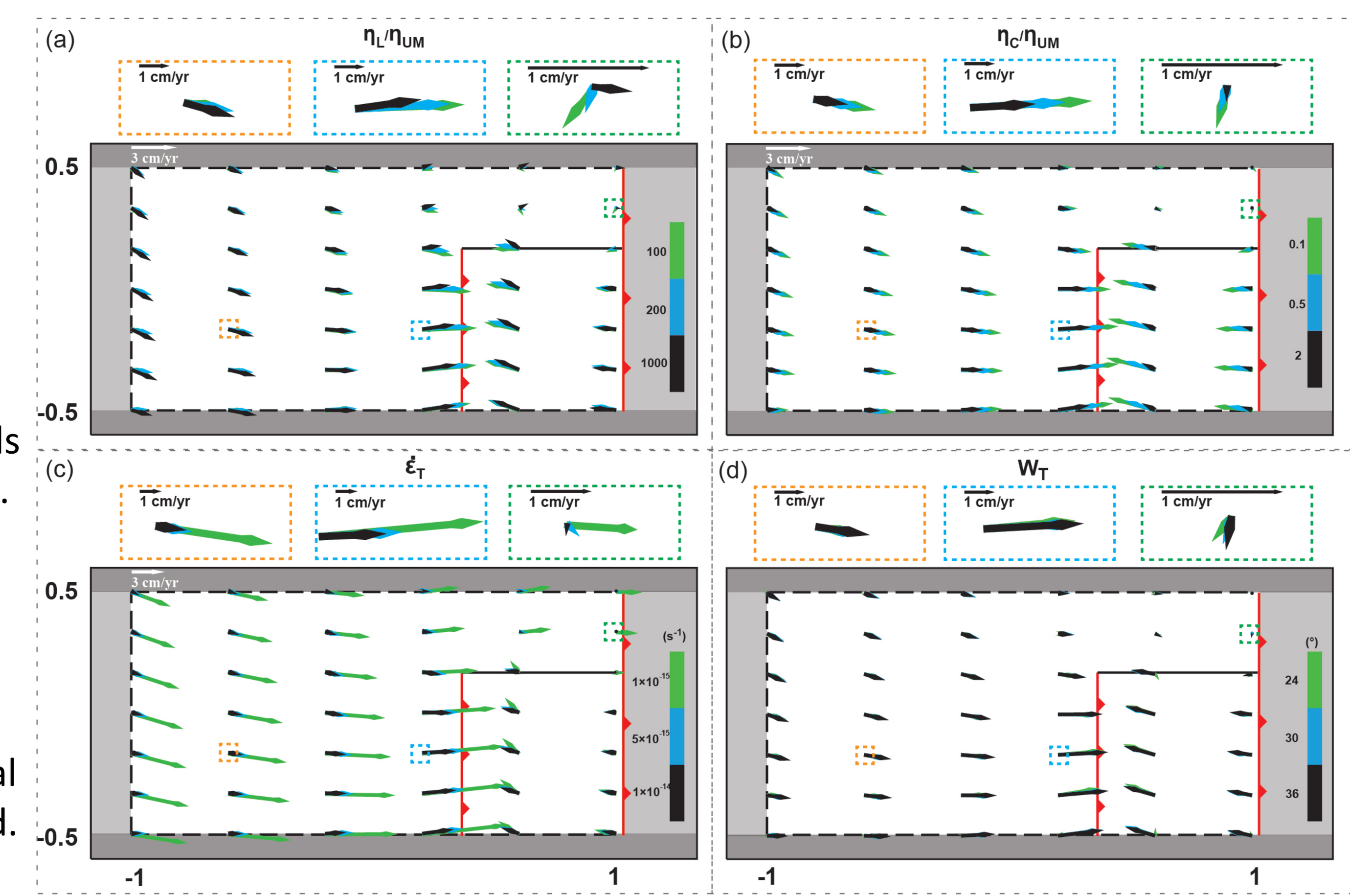


Plate velocity changes for variable subduction parameters

- We examine the impacts of variable lithospheric and crustal strengths: η_L/η_m (100, 1000), η_C/η_m (0.1, 2). We also vary plate width, W_T (24°, 36°), and test the impact of power-law flow, $\dot{\varepsilon}_T$ (10^{-15}, 5×10^{-15}, 10^{-14} s^{-1})
- In all models, double subduction drives faster plate speeds towards the trench. Overall, for lower η_L/η_m, η_C/η_m, or $\dot{\varepsilon}_T$ (upper panels), the magnitude ratio is higher.

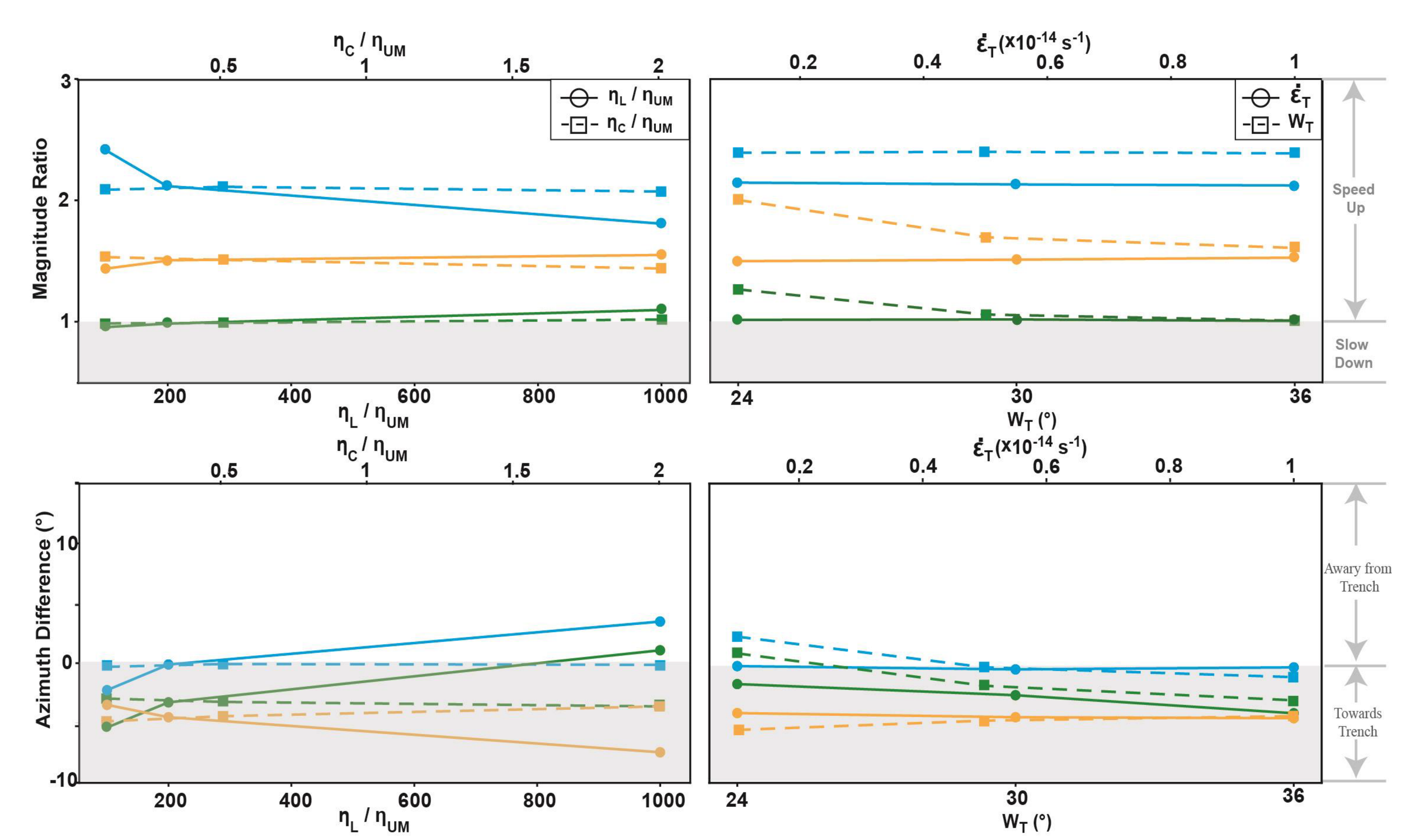


- The right-hand figure displays the vector change of plate velocities in the double slab model relative to the single slab case:
- The new subduction zones rotates the main plate towards the double subduction region. The weaker the lithosphere, crust, or mantle, the greater this rotation.
- To the side of the new subduction zone, only minimal velocity changes are observed.



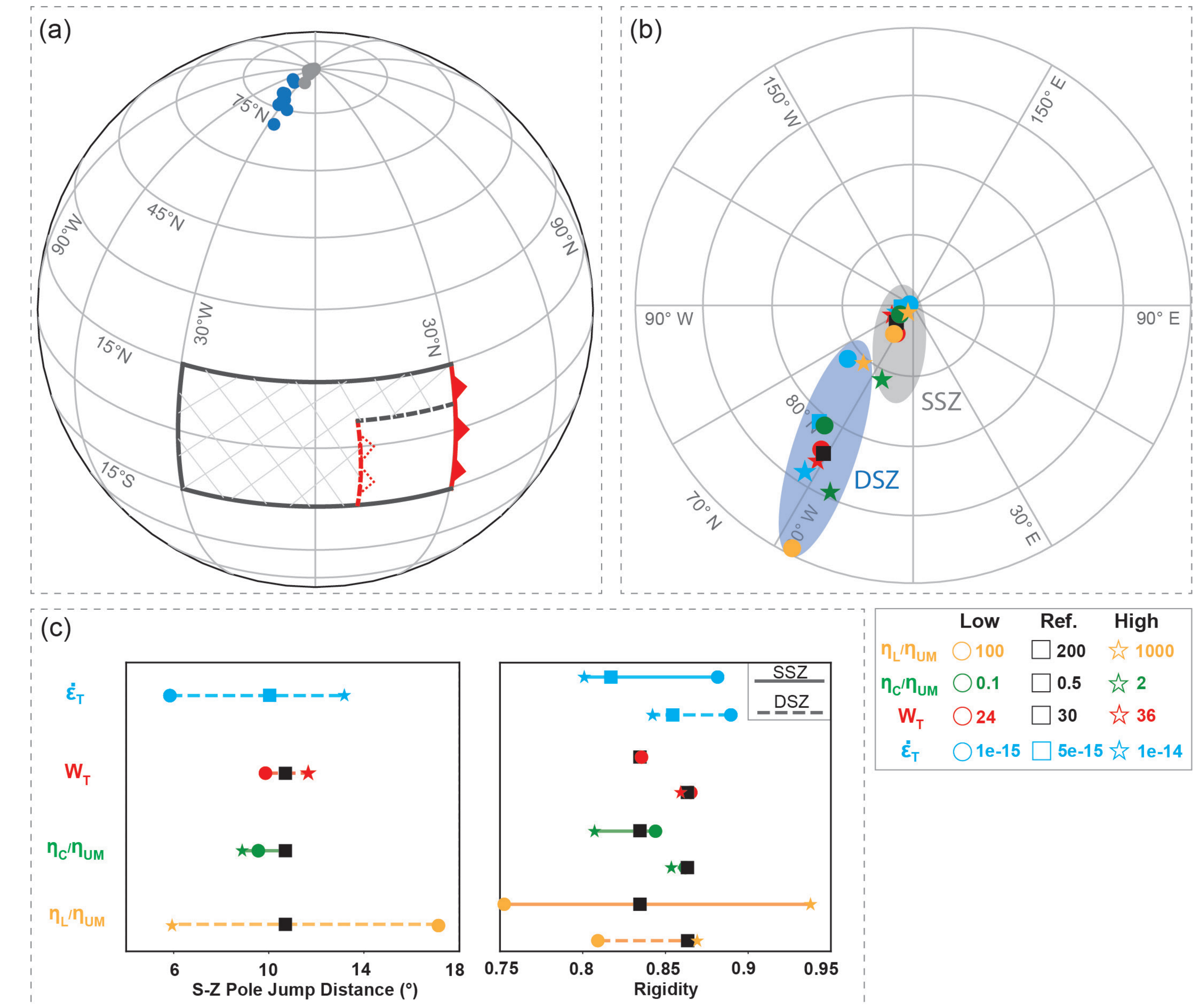
Quantifying the plate velocity changes

- Generally, plate velocities increase by a factor of 1.5 (orange curves). This aligns with the Africa Plate velocity surge, from ~45 to 70-80 mm/yr [5].
- Far-field plate rotation of 3-8° is observed. This is much lower than the 30-35° Pacific rotation, and in the opposite direction to Africa Plate rotation.



Subducting Plate's Euler Pole

- We examine the locations of the best-fitting Euler Poles of the main plate (shaded area in panel, with and without double subduction (a). Double subduction shifts the Euler Pole locations by great circle distances of 6-17° (b,c). This is comparable to ~6° in Africa Plate case[5] and ~10° in Pacific Plate case[13-16].
- We also calculate the plate rigidity (1 = perfectly rigid plate). Most rigidities exceed 0.8, as is in line with previous studies[11,12].



Conclusions

- For all mechanical and geometrical parameters, intra-plate subduction induces an overall plate speed-up with an average ~50% increase. Plate velocity direction changes are < 10°. The impact is higher for weaker lithosphere, crust or (power-law) mantle.
- Modeled plate rigidities (> 0.8) align with estimates from previous studies. For relatively rigid plates, double subduction induces an Euler Pole shift of by 6–14°.
- Our idealized models provide insight into the large-scale plate velocity changes associated with a 2nd subduction zone. While certain aspects of the primary geological examples align with our modeled behavior (e.g., the African plate speed-up; average shifts in Euler pole locations), others do not (e.g., the magnitude of Pacific Plate rotation) and therefore indicate a prominent role for additional processes.

Acknowledgements and References

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